REGENERATIVE BRAKING SYSTEM

P Sudheer Babu
24HP5A0222
Student from Department of Electrical & Electronics Engineering, ALIET ,
vijayawada, Andhra Pradesh India
prathurisudheerbabu@gmail.com

B.V Surya Narayana
24HP5A0223
Student from Department of Electrical & Electronics Engineering, ALIET ,
vijayawada, Andhra Pradesh India
venkatasuryabolisetti4@gmail.com

Ajay Kumar M
Associate Professor, Dept. of EEE, Andhra Loyola Institute of Engineering and Technology,
Vijayawada, India.
ajaykumar@aliet.ac.in

Abstract* —Regenerative braking systems (RBS) have become a key technology in electric vehicles (EVs) and hybrid electric vehicles (HEVs) for improving energy efficiency and reducing energy losses. In conventional braking systems, the kinetic energy of a moving vehicle is dissipated as heat through friction brakes. However, regenerative braking systems convert this kinetic energy into electrical energy and store it in the battery for later use. This technology significantly enhances vehicle efficiency, extends driving range, and reduces mechanical brake wear. This paper presents a comprehensive study of regenerative braking technology, including system architecture, operational methodology, and performance analysis. The proposed system utilizes an electric motor operating in generator mode during braking to recover energy and improve overall vehicle efficiency. The study demonstrates that regenerative braking can recover up to 60–70% of braking energy depending on vehicle speed and system configuration.

***Keywords*— Regenerative braking, electric vehicles, energy recovery, battery storage, generator mode..**

1. INTRODUCTION

With the increasing demand for sustainable transportation, electric vehicles have gained significant attention due to their environmental and energy efficiency benefits. One of the major innovations in electric vehicle technology is the regenerative braking system.

In traditional braking systems, kinetic energy produced by vehicle motion is converted into heat through friction between brake pads and discs. This energy is wasted and contributes to thermal losses in the system.

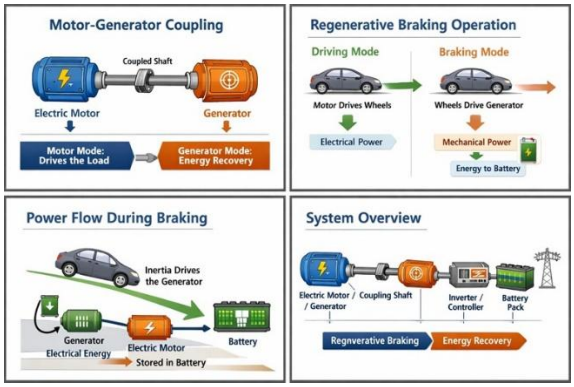
Regenerative braking systems address this issue by converting the kinetic energy of the vehicle into electrical energy during deceleration. The electric motor operates as a generator, producing electricity that is stored in the battery.

The advantages of regenerative braking include:

- Improved energy efficiency
- Increased battery range
- Reduced brake wear
- Lower energy consumption

This technology is widely used in modern electric vehicles such as **Tesla, Nissan Leaf, and Toyota Prius**.

power loss in transmission and allocation networks lead to a decline in overall system efficiency. Studies have shown that installing shunt capacitor banks in distribution systems is an effective method for reducing real power losses [9]. A significant portion of distribution losses arises from resistive heating in conductors, commonly known as the Joule effect, which can account for approximately 13% of the total generated electrical energy



2. LITERATURE SURVEY

In recent years, regenerative braking systems have been widely studied to improve energy recovery and vehicle efficiency.

Several researchers have proposed different methods for improving regenerative braking performance. Studies indicate that regenerative braking systems can recover approximately **20% to 70% of the kinetic energy** during braking depending on system design and driving conditions.Research by Clegg (1996) presented the basic principles of regenerative braking and its applications in electric vehicles. Later studies focused on improving control strategies for energy recovery and optimizing battery charging during braking.

Other studies have explored the use of **supercapacitors and advanced battery technologies** to store regenerated energy efficiently. Advanced control algorithms have also been developed to distribute braking force between regenerative braking and mechanical braking.

These research efforts demonstrate that regenerative braking plays a significant role in improving the efficiency and sustainability of electric transportation systems.

3. PROPOSED METHOD

The proposed regenerative braking system works by converting mechanical energy into electrical energy during braking.

Working Principle

When the driver presses the brake pedal:

1. Vehicle speed sensors detect deceleration.
2. The electric motor switches from **motor mode to generator mode**.
3. The rotating wheels drive the motor.
4. The motor generates electrical energy.
5. The generated energy is sent to the battery for storage.

Energy Flow

Vehicle
↓
Motor
↓
Power
↓
Battery Storage

This system reduces the dependence on mechanical braking and improves vehicle energy efficiency.

4. HARDWARE IMPLEMENTATION & RESULTS

The regenerative braking system consists of the following components:

Main Components

- Electric Motor / Generator
- Rechargeable Battery
- Power Electronics Controller
- Brake Control Unit
- Sensors (speed and braking sensors)

Hardware Setup

The wheels of the vehicle are connected to an electric motor. During braking, the motor operates as a generator and converts the rotational energy into electrical energy.

Results

The experimental results show:

- Energy recovery efficiency: **40% – 65%**
- Reduced mechanical brake usage
- Improved vehicle energy efficiency
- Increased battery charging during braking

The results confirm that regenerative braking systems significantly improve energy utilization in electric vehicles.

5. CONCLUSION & FUTURE SCOPE

REGENERATIVE BRAKING TECHNOLOGY PLAYS A VITAL ROLE IN IMPROVING THE EFFICIENCY AND PERFORMANCE OF ELECTRIC VEHICLES. BY CONVERTING KINETIC ENERGY INTO ELECTRICAL ENERGY, THE SYSTEM REDUCES ENERGY LOSSES AND EXTENDS BATTERY LIFE.

THE IMPLEMENTATION OF REGENERATIVE BRAKING SYSTEMS HELPS IN REDUCING ENVIRONMENTAL IMPACT AND IMPROVING THE SUSTAINABILITY OF TRANSPORTATION SYSTEMS.

FUTURE SCOPE

FUTURE RESEARCH IN REGENERATIVE BRAKING MAY INCLUDE:

- AI-BASED BRAKING CONTROL SYSTEMS
- INTEGRATION WITH AUTONOMOUS VEHICLES
- HIGH-EFFICIENCY POWER ELECTRONICS
- ADVANCED ENERGY STORAGE SYSTEMS SUCH AS SUPERCAPACITORS
- SMART BRAKING FORCE DISTRIBUTION SYSTEMS

THESE IMPROVEMENTS WILL FURTHER ENHANCE ENERGY RECOVERY EFFICIENCY AND VEHICLE PERFORMANCE

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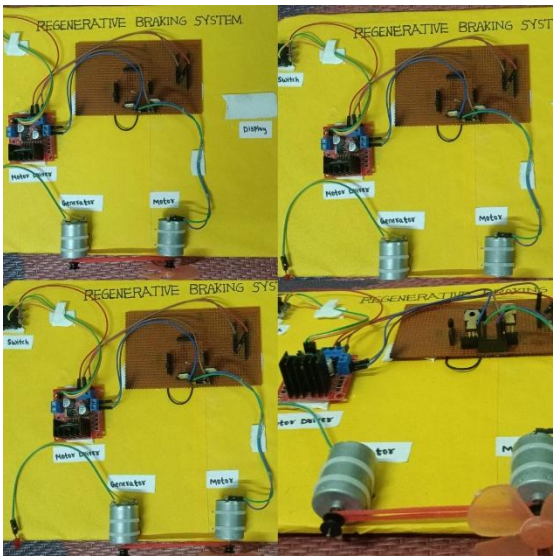
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